PROBLEM 07

Skater Towed by a Vehicle: Rotating Frame

Original statement · Problems.pdf, p. 8

Rotating Frame + Inertia Forces

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

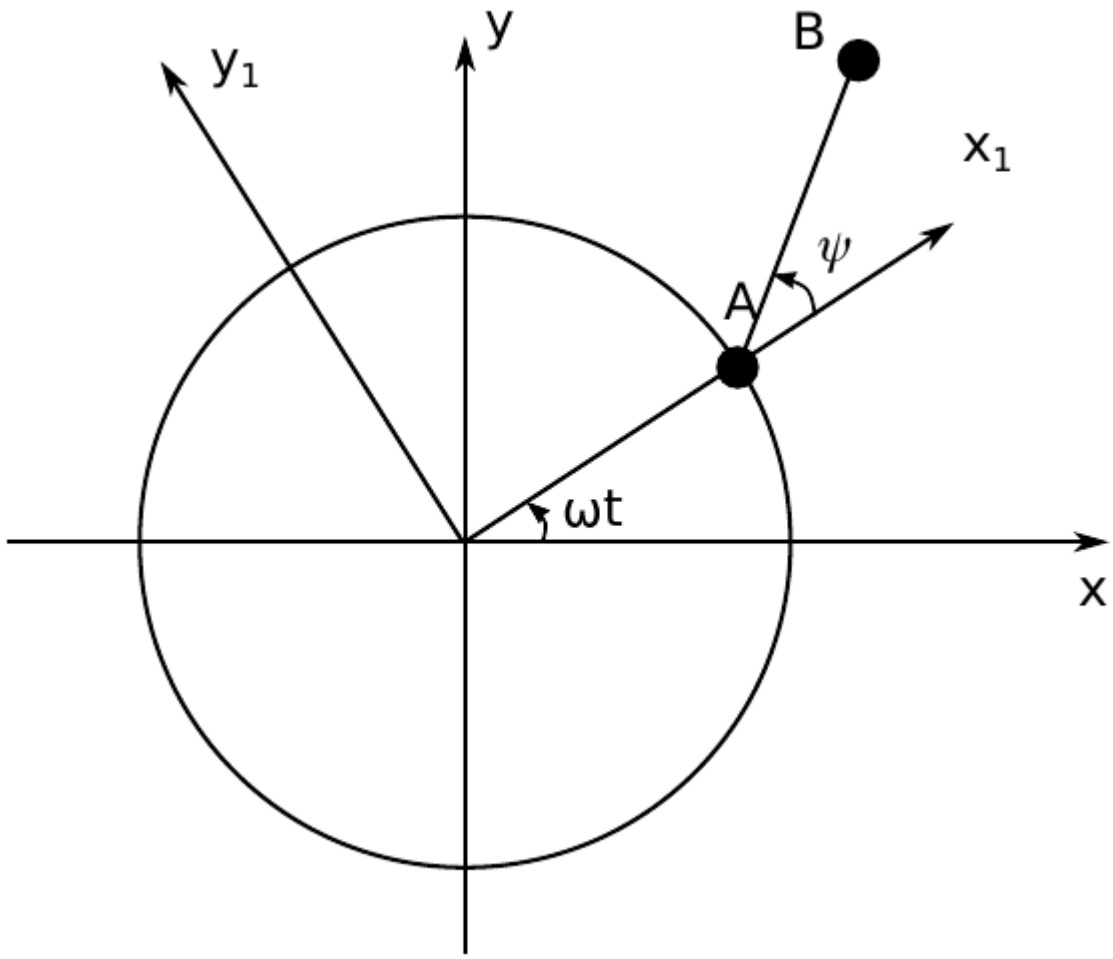


Figure: Inertial axes Oxy, rotating axes Ox1y1 turned by ωt, vehicle A on the circle of radius a, bar AB at angle ψ from Ox1, skater B (Problems.pdf, p. 8).

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [\dots]\, d\psi = \int [\dots]\, dt$ (consider that $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\, dt$).

(d) Determine the tension on the bar.

(e)

If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **Physical system:** the vehicle A moves on the circle of radius a centred at O with constant angular rate ω. The skater B (mass m) is joined to A by a massless rigid bar of length $l = a$ and slides on a smooth horizontal plane.
- **Why a rotating frame.** In the frame Ox_1y_1 that turns with the line OA, the vehicle is at rest at $(a, 0)$, so the skater moves on a fixed circle of radius a around A: one coordinate ψ describes the motion and the constraint is fixed in that frame. The price is that Ox_1y_1 is non-inertial, so we must add the inertia forces of Notes Eq. 4.62 to Newton's law (Notes Eq. 4.63).
- **Hypotheses:**
  1. Oxy is inertial; the plane is smooth (its reaction is vertical and cancels the weight, so neither enters the horizontal dynamics).
  2. The bar is massless, so the force it exerts on B is directed along the bar: $-T\,\mathbf{u}_{AB}$, with $T > 0$ meaning tension.
  3. ω is constant.
- **Degrees of freedom:** B moves in the plane (2 coordinates) with one constraint $|AB| = a$: **1 DOF**, the angle ψ.
- **Initial conditions:** A at $(a, 0)$ and B at rest at $(2a, 0)$, so $\psi(0) = 0$ and $\mathbf{v}_0^B(0) = \mathbf{0}$ (this gives $\dot{\psi}(0)$ in part (c)).

Symbol	Meaning	SI unit
m	mass of the skater B	kg
a	radius of the vehicle's circle and bar length	m
ω	angular rate of the vehicle (constant)	rad/s
ψ	angle from Ox_1 to the bar AB	rad
T	tension of the bar (positive when pulling B toward A)	N

PHASE 2

Reference Frames & Vector Bases

- **Frames.** $S_0 = Oxy = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ (inertial) and $S_1 = Ox_1y_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ with $\mathbf{i}_1$ along OA. The origin of S_1 is O (it plays the role of the generic origin "A" of Notes Ch. 3, which here is the fixed point O; the letter A is kept for the vehicle as in the statement) and the particle is $P \equiv B$.
- **Attitude.** The vehicle starts on Ox and turns counter-clockwise, so the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is ωt. The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$; the columns are unit vectors ($\cos^2\omega t + \sin^2\omega t = 1$) and mutually orthogonal ($-\cos\omega t \sin\omega t + \sin\omega t \cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1.** Simple rotation about $\mathbf{k}_0 = \mathbf{k}_1$ (Slide 16):

$$\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\omega}\,\mathbf{k}_1 = \mathbf{0}, \qquad \mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$$

- **Bar basis.** $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$ (from A to B) and $\mathbf{u}_\perp = \mathbf{k}_1 \times \mathbf{u}_{AB} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$. Their matrix $\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}$ (completed with $\mathbf{k}_1$) is again a simple rotation, with $\det = 1$ and $RR^T = I$.

PHASE 3

Derivation

3.1 Kinematics

(a) Relative acceleration $\mathbf{a}_1^B$

> **Pedagogical Justification:** the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components are differentiated.

Relative position (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$): $\mathbf{r}_1^B = \mathbf{OA} + \mathbf{AB}$,

$$\mathbf{r}_1^B = a\,\mathbf{i}_1 + a\,(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1) = a(1+\cos\psi)\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

Relative velocity, with the chain rule $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot\psi$ and $\frac{d}{dt}\sin\psi = \cos\psi\,\dot\psi$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -a\dot\psi\sin\psi\,\mathbf{i}_1 + a\dot\psi\cos\psi\,\mathbf{j}_1 = a\dot\psi\,\mathbf{u}_\perp$$

Relative acceleration, with the product rule $\frac{d}{dt}(\dot\psi\sin\psi) = \ddot\psi\sin\psi + \dot\psi^2\cos\psi$ and $\frac{d}{dt}(\dot\psi\cos\psi) = \ddot\psi\cos\psi - \dot\psi^2\sin\psi$:

$$\boxed{\mathbf{a}_1^B = -a\left(\ddot\psi\sin\psi + \dot\psi^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot\psi\cos\psi - \dot\psi^2\sin\psi\right)\mathbf{j}_1 = a\ddot\psi\,\mathbf{u}_\perp - a\dot\psi^2\,\mathbf{u}_{AB}}$$

(the last form follows by grouping the $\ddot\psi$ and $\dot\psi^2$ terms: circular motion of radius a about A, tangential plus normal part).

(b) Centripetal and Coriolis terms

> **Pedagogical Justification:** they are the two non-zero rotation terms of Notes Eq. 3.27 (Slide 25); the Euler term vanishes because $\boldsymbol{\alpha}_{10} = \mathbf{0}$ and the origin term because $\mathbf{a}_0^O = \mathbf{0}$.

Cross products used: $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$.

Centripetal term.

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = \omega\mathbf{k}_1 \times [a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1] = a\omega(1+\cos\psi)\,\mathbf{j}_1 - a\omega\sin\psi\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = \omega\mathbf{k}_1 \times [-a\omega\sin\psi\,\mathbf{i}_1 + a\omega(1+\cos\psi)\,\mathbf{j}_1] = -a\omega^2\sin\psi\,\mathbf{j}_1 - a\omega^2(1+\cos\psi)\,\mathbf{i}_1$$

$$\boxed{\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2 \left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right] = -\omega^2\,\mathbf{r}_1^B}$$

It equals $-\omega^2 \mathbf{r}_\perp$ with $\mathbf{r}_\perp = \mathbf{r}_1^B$ because $\mathbf{r}_1^B$ lies in the plane normal to $\boldsymbol{\omega}_{10}$ (Slide 30).

Coriolis term.

$$2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = 2\omega \mathbf{k}_1 \times \left(-a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\sin\psi\,\mathbf{j}_1 - 2a\omega\dot{\psi}\cos\psi\,\mathbf{i}_1$$

$$\boxed{2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = -2a\omega\dot{\psi}\left(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\,\mathbf{u}_{AB}}$$

Both agree with the official key (Problems.pdf, p. 72), where the centripetal term is called "centrifugal acceleration".

Absolute acceleration (Notes Eq. 3.27):

$$\mathbf{a}_0^B = \underbrace{\mathbf{a}_0^O}_{\mathbf{0}} + \mathbf{a}_1^B + \underbrace{\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^B}_{\mathbf{0}} - \omega^2 \mathbf{r}_1^B - 2a\omega\dot{\psi}\,\mathbf{u}_{AB}$$

3.2 Forces

(c) Forces on the skater in S_1

> **Pedagogical Justification:** in S_1 Newton's law holds with the inertia forces (Notes Eqs. 4.62–4.63): $m\mathbf{a}_1^B = \mathbf{F} + \mathbf{F}_i$. Here $\mathbf{F} = -T\,\mathbf{u}_{AB}$ (horizontal part), $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = \mathbf{0}$, and

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = ma\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right], \qquad \mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = 2ma\omega\dot{\psi}\,\mathbf{u}_{AB}$$

3.3 Equations of Motion

(c, continued) Equation of motion

The unknown T disappears if we project on $\mathbf{u}_\perp$, the only direction along which the bar exerts no force.

Projections on $\mathbf{u}_\perp = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B \cdot \mathbf{u}_\perp = ma\ddot{\psi}$ (from the grouped form of part (a), since $\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = 0$);
- $-T\,\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = 0$ and $\mathbf{F}_{COR} \cdot \mathbf{u}_\perp = 0$;
- $\mathbf{F}_C \cdot \mathbf{u}_\perp = ma\omega^2\left[-(1+\cos\psi)\sin\psi + \sin\psi\cos\psi\right] = -ma\omega^2\sin\psi$.

$$ma\ddot{\psi} = -ma\omega^2\sin\psi \quad\Longrightarrow\quad \boxed{\ddot{\psi} + \omega^2\sin\psi = 0}$$

In S_1 the skater behaves as a pendulum of length a hanging from A: about A the centrifugal force $m\omega^2(\mathbf{OA} + \mathbf{AB})$ has the same moment as the uniform field $ma\omega^2\,\mathbf{i}_1$ (because $\mathbf{AB} \times \mathbf{AB} = \mathbf{0}$), so the "effective gravity" has strength $a\omega^2$ and points along $+\mathbf{i}_1$.

3.4 Integration and Answers

(c, continued) First integral, initial rate and quadrature

First integral (hint of the statement). Multiply by $2\dot{\psi}$. By the chain rule $\frac{d}{dt}\left(\dot{\psi}^2\right) = 2\dot{\psi}\ddot{\psi}$ and $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot{\psi}$, so

$$2\dot{\psi}\ddot{\psi} = -2\omega^2\sin\psi\,\dot{\psi} \quad\Longleftrightarrow\quad \frac{d}{dt}\left(\dot{\psi}^2\right) = 2\omega^2\frac{d}{dt}\left(\cos\psi\right)$$

Integrate from 0 to t and apply Barrow's rule $[F]_0^t = F(t) - F(0)$:

$$\dot{\psi}^2(t) - \dot{\psi}^2(0) = \int_0^t 2\dot{\psi}\ddot{\psi}\,dt' = 2\omega^2\left[\cos\psi\right]_{\psi(0)}^{\psi(t)} = 2\omega^2\left(\cos\psi - 1\right)$$

Initial rate. By Notes Eq. 3.26, $\mathbf{v}_0^B = \mathbf{v}_0^O + \mathbf{v}_1^B + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^B$. At $t = 0$: $\psi = 0$, $\mathbf{r}_1^B = 2a\,\mathbf{i}_1$, $\mathbf{v}_1^B = a\dot{\psi}(0)\,\mathbf{j}_1$, $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = 2a\omega\,\mathbf{j}_1$, and $\mathbf{v}_0^B = \mathbf{0}$:

$$\mathbf{0} = \Big[a\dot\psi(0) + 2a\omega\Big]\,\mathbf{j}_1 \quad\Longrightarrow\quad \dot\psi(0) = -2\omega$$

Hence, with $1+\cos\psi = 2\cos^2(\psi/2)$,

$$\dot\psi^2 = 4\omega^2 + 2\omega^2(\cos\psi - 1) = 2\omega^2(1+\cos\psi) = 4\omega^2\cos^2\frac{\psi}{2}$$

For $-\pi < \psi < \pi$ the factor $\cos(\psi/2) > 0$, so $\dot\psi$ cannot vanish nor change sign there; $\dot\psi(0) = -2\omega < 0$ selects

$$\boxed{\dot\psi = -2\omega\cos\frac{\psi}{2}, \qquad -\pi < \psi \le 0}$$

which agrees with the official key (Problems.pdf, p. 72).

Energy check in S_1 (Slides 29–30). The origin is fixed and $\boldsymbol{\omega}_{10}$ is constant, so $\mathbf{F}_C$ has the potential $V_1 = -\tfrac12 m\omega^2 r_\perp^2$, $\mathbf{F}_{COR}$ does no work, and the bar force does no work in S_1 ($-T\mathbf{u}_{AB}\cdot a\dot\psi\mathbf{u}_\perp = 0$). With $r_\perp^2 = a^2\left[(1+\cos\psi)^2 + \sin^2\psi\right] = 2a^2(1+\cos\psi)$:

$$E_1^B = \frac12 ma^2\dot\psi^2 - ma^2\omega^2(1+\cos\psi) = \text{const} = \frac12 ma^2(4\omega^2) - 2ma^2\omega^2 = 0$$

which gives again $\dot\psi^2 = 2\omega^2(1+\cos\psi)$.

Quadrature. Separate variables in $\frac{d\psi}{dt} = -2\omega\cos\frac{\psi}{2}$:

$$\boxed{\int_0^{\psi}\frac{d\psi'}{2\cos(\psi'/2)} = -\int_0^t \omega\,dt'}$$

The left integral is elementary. Substitute $w = \psi'/2$, $d\psi' = 2\,dw$, limits $0 \to \psi/2$; the antiderivative of $\sec w$ is $\ln\left|\sec w + \tan w\right|$ (its derivative is $\frac{\sec w\tan w + \sec^2 w}{\sec w + \tan w} = \sec w$). By Barrow's rule:

$$\int_0^{\psi/2} \frac{dw}{\cos w} = \left[\ln\left|\sec w + \tan w\right|\right]_0^{\psi/2} = \ln\left|\sec\frac{\psi}{2} + \tan\frac{\psi}{2}\right| - \ln 1 = -\omega t$$

so $\psi(t)$ is given implicitly by $\sec\frac{\psi}{2} + \tan\frac{\psi}{2} = e^{-\omega t}$; as $t \to \infty$, $\psi \to -\pi$, i.e. B tends to the centre O ($\mathbf{r}_1^B \to \mathbf{0}$) without ever reaching it.

(d) Tension on the bar

> **Pedagogical Justification:** project $m\mathbf{a}_1^B = -T\mathbf{u}_{AB} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathbf{u}_{AB}$, the direction in which the bar force acts.

Projections on $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B \cdot \mathbf{u}_{AB} = -ma\dot{\psi}^2$;
- $\mathbf{F}_C \cdot \mathbf{u}_{AB} = ma\omega^2\left[(1+\cos\psi)\cos\psi + \sin^2\psi\right] = ma\omega^2(1+\cos\psi)$;
- $\mathbf{F}_{COR} \cdot \mathbf{u}_{AB} = 2ma\omega\dot{\psi}$.

$$-ma\dot{\psi}^2 = -T + ma\omega^2(1+\cos\psi) + 2ma\omega\dot{\psi} \quad\Longrightarrow\quad T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$$

Substitute the first integral with $c \equiv \cos(\psi/2)$: $\dot{\psi}^2 = 4\omega^2 c^2$, $\dot{\psi} = -2\omega c$, $1+\cos\psi = 2c^2$:

$$T = ma\omega^2\left(4c^2 - 4c + 2c^2\right) = ma\omega^2\left(6c^2 - 4c\right)$$

$$\boxed{T = 2ma\omega^2\cos\frac{\psi}{2}\left(3\cos\frac{\psi}{2} - 2\right)}$$

which agrees with the official key (Problems.pdf, p. 72). At $t = 0$: $T(0) = 2ma\omega^2 > 0$.

(e) Does a string become loose?

> **Pedagogical Justification:** a string can only pull, so it stays taut while $T \geq 0$; it goes slack at the first instant where T would become negative.

As ψ decreases from 0 toward $-\pi$, $c = \cos(\psi/2)$ decreases monotonically from 1 toward 0. The factor c stays positive, and $3c - 2$ changes sign at $c = 2/3$:

$$\cos\frac{\psi}{2} = \frac{2}{3} \quad \Longrightarrow \quad \boxed{\psi_{\text{loose}} = -2\arccos\frac{2}{3} \approx -1.682 \text{ rad} \approx -96.38^\circ}$$

(official key: $\psi = -2\arccos(2/3)$). **When:** at $\psi/2 = -\arccos(2/3)$, $\sec(\psi/2) = 3/2$ and $\tan(\psi/2) = -\sqrt{1 - 4/9}/(2/3) = -\sqrt{5}/2$, so from part (c)

$$\omega t_{\text{loose}} = -\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right) = \ln\frac{2}{3 - \sqrt{5}} = \ln\frac{3 + \sqrt{5}}{2} \approx 0.9624, \qquad t_{\text{loose}} \approx \frac{0.962}{\omega}$$

So yes: the string goes slack after about $0.96/\omega$ seconds, when the bar has turned 96.4° clockwise from Ox_1.

PHASE 4

Interpretation & Checks

- **Reading in S_1.** The rotating observer sees B as a pendulum of length a hung from A in an effective field of strength $a\omega^2$ pointing away from O (the centrifugal force), whose "lowest" point is $\psi = 0$. The skater starts there with relative speed $2a\omega$, which is exactly the separatrix energy ($E_1^B = 0$): it creeps toward the "inverted" position $\psi = -\pi$ (where B would sit at O) in infinite time.
- **Why T drops.** The Coriolis force $2ma\omega\dot\psi\,\mathbf{u}_{AB}$ points toward A when $\dot\psi < 0$, and the centrifugal pull along the bar, $ma\omega^2(1 + \cos\psi)$, shrinks as B swings in; when the required centripetal force $ma\dot\psi^2$ is supplied by these inertia forces alone the bar is unloaded ($T = 0$).
- **Limit $\omega \to 0$:** $\dot\psi \to 0, T \to 0$ (nothing moves).
- **Dimensions:** $[\mathbf{a}_1^B] = [a\ddot\psi] = [a\dot\psi^2] = \mathrm{m\,s^{-2}}$; $[\omega^2 a] = [\omega\dot\psi a] = \mathrm{m\,s^{-2}}$; $[T] = [ma\omega^2] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$; $[\omega t] = 1$.

- **Verification (sympy):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equals $\mathbf{a}_1^B - \omega^2\mathbf{r}_1^B - 2a\omega\dot{\psi}\mathbf{u}_{AB}$; the EOM, $\dot{\psi}(0) = -2\omega$, $dE_1^B/dt = 0$ along the EOM, the tension and ψ_{loose} are reproduced; an RK4 integration with $\omega = 1$ reaches $\cos(\psi/2) = 2/3$ at $t = 0.9625$ s, matching $\ln\frac{3+\sqrt{5}}{2} = 0.9624$.
- **Check against the official key:** the key of `Problems.pdf` (p. 72) gives the centripetal and Coriolis terms, $\dot{\psi} = -2\omega\cos(\psi/2)$, $T = 2ma\omega^2\cos(\psi/2)\,(3\cos(\psi/2) - 2)$ and $\psi = -2\arccos(2/3)$ for the string going slack. All of them coincide with the results above (no discrepancy).

Source: Problems.pdf (PDF page 8); key on PDF page 72